

Final Project Draft 2 - Girls with Books

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Proposed Site URL: <https://www.girlswithbooks.org/home>

Site Description:

- Girls With Books is a site belonging to the merging of two small organizations: Empowerment Through Education Foundation and Project Education South Sudan. Their goal is to educate South Sudanese girls and young women to become peacemakers and positive change-makers. They provide education for primary and secondary students, along with post-graduation programs, and pay tuition for girls to attend private schools.
- The site offers extensive details on what the program provides, the crucial problems being addressed, and pages for contacting them as well as donating to their cause.

Why We Chose This Site:

- We chose this site because we like their message and what they stand for. They help underserved communities outside of the United States and everyone deserves the opportunity to read. Their site is simple and gets their message across, but we believe it can be improved. Non profit organizations need all the outreach they can get and an easily navigated website will keep people interested.

Brief List of Accessibility/Usability Issues Initially Identified:

- 1.4.4 Resize Text: When the site's text is increased to 200%, content/functionality is lost. The page requires scrolling horizontally to read all information.
- 1.1.1 Non-Text Content: Alternative text for images is not useful (e.g., alt="IMG_9842.jpg").
- 2.4.6 Headings and Labels; 2.4.10 Section Headings: The site does not adhere to the heading hierarchy, skipping levels, and does not mark all section headings as such.

Planned Site Owner to Interview:

- We have not been able to obtain cooperation from the owner or someone from the organization, so we will be doing a second usability test with someone who uses assistive technology.

Planned Remote User Test:

- Yes, we will be conducting a remote user test with someone who has color blindness and dyslexia who often has an issue with accessing digital material.

Project Goals:

With our redesign, we are going to:

- Correct low color contrast issues
 - Between text and background
- Improve keyboard navigation
 - On the donation form, the radio checkbox for payment method does not work for selecting the type using just a keyboard, losing the keyboard focus and ability to select a choice
- Improve the organization and structure of the site
 - The site currently has a very clashing background image on a few pages that make the content harder to read and also creates chaos on the page
 - The density of the page is a lot right now, making it overwhelming to look at
 - There lacks a heading hierarchy, with the pages only using h3
- Add context to the site
 - Create meaningful alternative text for the images and links, as they are missing or not useful as of now (e.g., alt="IMG_9842.jpg")

Heuristic Evaluation:

<u>Section</u>	<u>Item</u>	<u>Heuristic Question</u>	<u>Your Findings</u>	<u>Recommendation and WCAG 2.1 Language</u>
Page Title	Page Title	Does each page have a unique/descriptive title?	The page titles for most of the pages are descriptive, but there are two pages with the same title and should be more descriptive	I would change the home page title from “Girls With Books Girls’ Education South Sudan” to “Home Girls With Books” and for the other page to switch the order. This is WCAG 2.1 Guideline 2.4 - Navigable: 2.4.2 Page Titled A
Media	Alt Text for Images	Do all images have appropriate alt text which describes the content and function?	The images on the site have very nonhelpful alt text such as “IMG_9842.jpg”	I would change the alt texts for the images to describe the content rather than a jpg name. This is Guideline 1,1 Text Alternatives: 1.1.1 Non-text Content A
Design and Layout	Color Contrast	Does the foreground/background and contrast meet WCAG guidelines?	There are a few low contrast errors between text and the background, as well as between some foreground and backgrounds as well	Our solution would be to change the colors of the text to meet the 4.5:1 contrast ratio for Guideline 1.4 - Distinguishable: 1.4.3 (minimum) AA in order to make the content accessible to read
Text	Resize Text	Can users resize text without loss of content?	The text enlarges when the zoom is increased on the page, but once it reaches 200%, then the	We would add flex or styling rules to adjust for the zoom without losing content. This is WCAG 2.1 Guideline: 1.4.4

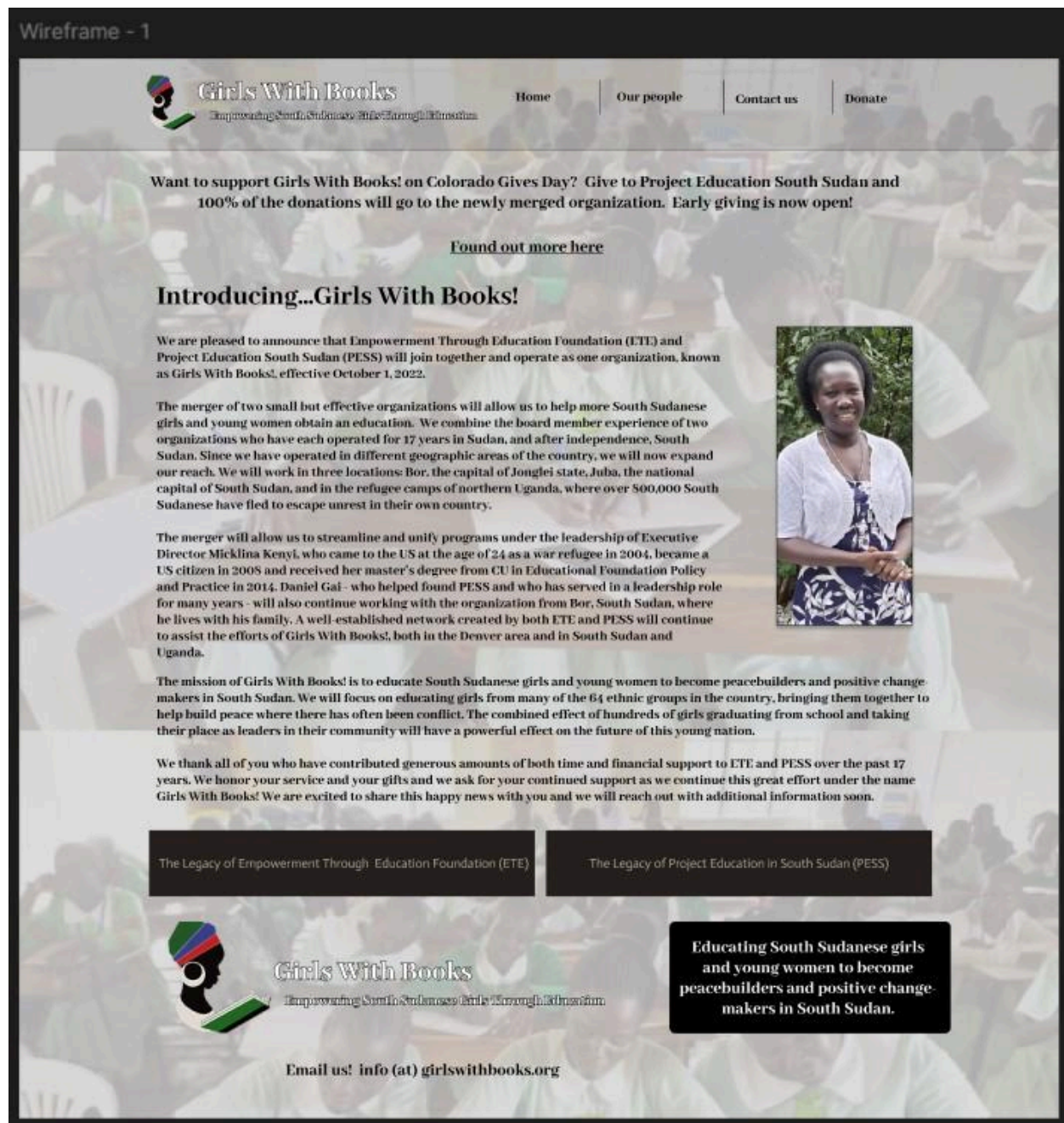
			content must use a horizontal scroll	Resize Text
Navigation	Keyboard Accessibility	Are all the interactive elements accessible by the keyboard and can the user navigate the site and interact with all features through only the keyboard?	Most of the content is accessible through the keyboard, the only exception being a form that is linked where the radios of a section are not able to be selected	We would fix the form to allow for complete keyboard accessibility in that field. This is the WCAG Guideline 2.1 - Keyboard Accessible: 2.1.1 Keyboard A
Navigation	Visual Focus Indicator	Is visual feedback provided for keyboard focus? In others, you can see your current location when you tab through the page.	Yes there is a keyboard focus that is a blue highlight around the selected link on the page.	We would maybe make the focus a bigger contrast color. This is WCAG Guideline 2.4 - Navigable: 2.4.7. Focus Visible AA. My only suggestion
Structure	Headings	Are there headings on the page, not just text that looks like a heading, but that is marked up as a heading: i.e. <h1>, <h2>, etc.?	The page uses only h3 headings, so there is not a followed hierarchy	I would change the h3 to h1 and then add lower ones if necessary for different sections. This is WCAG 2.1 Guideline 2.4 - Navigable: 2.4.6 Headings and Labels AA
Forms	Form Labels	Do all fields have associated labels?	Yes, all form fields have an associated label	This follows the WCAG 2.1 Guideline 3.3.2 Labels or Instructions
Structure	Content Organization	Is content organized in a logical way?	No, the homepage has large amounts of text without	I would add clearer section breaks, spacing, and headings to improve

			visual separation between sections	readability. This is WCAG 2.1 Guideline 1.3.1 Info and Relationships A.
Navigation	Clarity	Is it easy to understand where to go next on the site?	No, navigation options are limited and do not always clearly guide users to key actions.	I would improve navigation labels and add clear actions. This is WCAG 2.1 Guideline 2.4.6 Headings and Labels AA.
Interaction	Links and Bittons	Are interactive elements clearly identifiable?	No, some clickable text links are not distinct enough from regular text.	I would style links with stronger visual indicators. This is WCAG 2.1 Guideline 1.4.1 Use of Color
Forms	Assistance	Are users given helpful instructions when filling out forms?	The donate form is set up well, but the contact form lacks guidance or strong error cues.	I would add better placeholder examples. This is WCAG 2.1 Guideline 3.3.2 Labels or Instructions A.
Structure	Consistency	Is the layout consistent across pages?	The overall branding is consistent, but some pages vary in structure and spacing.	I would standardize layouts across all pages. This is WCAG 2.1 Guideline 3.2.3 Consistent Navigation AA
Content	Language Simplicity	Is the language easy to understand for a broad audience?	No, some sections use long paragraphs that may be difficult for users with cognitive disabilities.	I would simplify language and break content into shorter sections. This is WCAG 2.1 Guideline 3.1.5 Reading Level AAA
Feedback and Errors	Error Identification	Are input errors detected and communicated to	The contact form does tell the user when an	We would add inline error messages for all form fields to

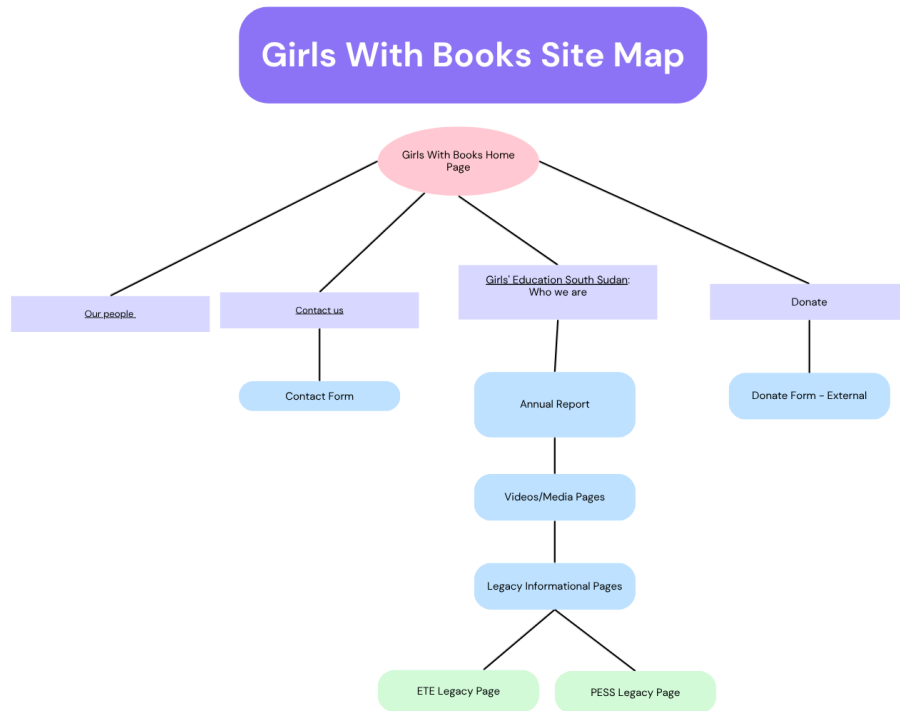
		the user?	invalid email is entered, but it does not give any feedback for the other form components.	communicate missing requirements and invalid formats. This follows WCAG 3.3 Input Assistance 3.3.1 Error Identification Level A.
Interaction	Links and Buttons	Can link purpose be understood independently of their surrounding context?	Some links require additional context to understand. For example, links with labels like “Click to learn more.”	We would write more descriptive link text that can be understood independently. This follows WCAG 2.4 Navigable 2.4.4 Link Purpose (In Context) Level AAA.
Interaction	Links and Buttons	Do links that open in new windows/tabs give a warning beforehand?	No, donation and links for Girls With Books! in the media do not warn the user that they will open in a new window.	We would include labels that alert users that these links open in a new window. This follows WCAG 3.2 Predictable 3.2.5 Change on Request Level AAA.
Navigation	Keyboard Accessibility	Can the webpage be navigated sequentially in a way that preserves meaning and operability?	On the home page, navigating through keyboard access puts the bypass block “Skip to Main Content” as the last option, which is not sequential.	We would change the navigation so the bypass block is shown after the nav bar when using the keyboard for navigation. This follows WCAG 2.4 Navigable 2.4.3 Focus Order Level A.
Aesthetic and Minimalist Design	Visual Design	Do unnecessary elements distract users from the information they	The background images do distract users from the main	We would remove the background images to make the content easier to scan

		really need?	information on the site.	and understand. This follows WCAG 1.4 Distinguishable 1.4.3 Contrast (Minimum) Level AA.
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Wireframe of Home Page: [Figma](#)



Site Map of Entire Site (up to 5 pages):



Persona Profile with Disabilities:

Bethany Poyton

I want to support girls' education and need to be able to find suitable organizations where I can donate and volunteer.

Bethany has always had a passion for education. As a middle school teacher for 28 years, she was able to feel fulfilled. Since retiring 2 years ago, she has been looking for ways to continue supporting young girls' education, either through donations or volunteer work.



When she discovers a potential organization to support, Bethany does her research to ensure that the organization is legitimate. She makes sure she fully understands their mission and outcomes before making a donation. She is more likely to trust organizations that have readable and inclusive websites.

Bethany began developing glaucoma symptoms 5 years ago. She first noticed slight blurry vision and headaches, but the symptoms have progressed significantly and she now experiences tunnel vision in her left eye and is extremely sensitive to light. She uses browser zoom and high-contrast mode when browsing on her computer. She appreciates when websites have clear headings, simple layouts, and keyboard-friendly navigation.

She uses her computer at her home, which she shares with her husband. She is comfortable using her home computer, but she can become confused when using newer computers that she is unfamiliar with.

Demographic Information

- 62 years old
- Located in Boulder, CO
- Married with 3 children
- Retired middle school teacher

Attitudes

- Passionate about girls' education.
- Frustrated by websites that use small text and are not clearly organized.

Goals

- To quickly understand an organization's mission and outcomes.

- To find accessible information about donations and volunteer opportunities.

Behaviors and Tasks

- She uses browser zoom to enlarge text before reading.
- She will quickly scan headings to find the information she is looking for.
- She will decide against supporting an organization if its website is too difficult to read or navigate because it makes the organization feel less trustworthy.

Skill Level and Environment

- She is comfortable with her own computer since she is familiar with its interface, but she struggles with newer technology where

Scenarios

- She visits the Girls with Books! website to learn about the organization and their mission. She will only stay on the site if the text is readable and the site is navigable at high zoom.
- She looks into donation information and expects the links to work well with keyboard navigation.

PROJECT DRAFT 2: Redesigned

Redesigned Wireframes [Figma](#)



Empowering South Sudanese Girls Through Education

Empowerment Through Education Foundation (ETF) and Project Education South Sudan (PESS) will join together and operate as one organization, known as Girls With Books!

Empowerment Through Education Foundation

The Empowerment Through Education Foundation works closely with schools, families and refugee camps to send girls to safe boarding schools where they can thrive.

Project Education South Sudan

Project Education South Sudan has been finding ways to fulfill our mission in the most changing environment of South Sudan.

Girls With Books merges two 17-year-old organizations that expand educational support for South Sudanese girls and young women across Somalia, India, and refugee camps in northern Uganda. We hope to broaden our geographic reach to serve the 500,000+ South Sudanese who have fled to Uganda.

Led by Executive Director Mikiisa Keny, a former war refugee turned US citizen, named her master's in Educational Foundation Policy and Practice from CU. Along side David Gat, a co-founder who will continue leading efforts on the ground in the, they will streamline the organizations' existing networks in Denver, South Sudan, and Uganda will support the newly unified Girls With Books!





Girls With Books!

[About](#) [Our People](#) [Contact](#) [Donate](#)



About Us

Girls With Books! was established on October 1, 2022, by two small and successful organizations, Empowerment Through Education Foundation and Project Education South Sudan, each with a 17-year history of serving the educational needs of South Sudanese youth.



What we do

We believe education is the most comprehensive and effective way to support the South Sudanese girls in child, teen, and refugee camps in northern Uganda. Our programs are in both primary and secondary education and provide real educational and living skills training to improve their future opportunities.

Because government-run public schools do not meet the quality needed to prepare students for higher employment and higher education, we serve both public and private schools. We also fund reading and leadership training programs, along with all educational supplies and materials, and provide incentives such as uniforms and books to girls who graduate.



Needs we address

South Sudan has one of the highest rates of "out of school" children in the world. According to UNESCO estimates 2021 – girls reach more in their lives. World Bank statistics show that 50 percent of the population is under 18 years old – 50% of women in the country can read and write, and less than 1% of girls eventually graduate from high school. Clearly, the work of Girls With Books! is profoundly needed in this, the world's newest nation.

Leadership



Girls With Books! is led by Executive Director Wanda Brown, a South Sudanese woman who fled Sudan in 2011 and arrived in the U.S. in 2012. She has a master's degree in education and has worked in education for over 20 years.

Ms. Brown has dedicated her life to helping young South Sudanese women access education on their way to become leaders in business, community, and government in their home countries. She oversees a team of teachers and volunteer mentors who provide local and personal support to 400 students in both public and private schools. She is supported by a dedicated board of directors, composed of women who are currently in South Sudan, as well as a team of volunteers who provide the support and resources needed to make this work possible.

Media

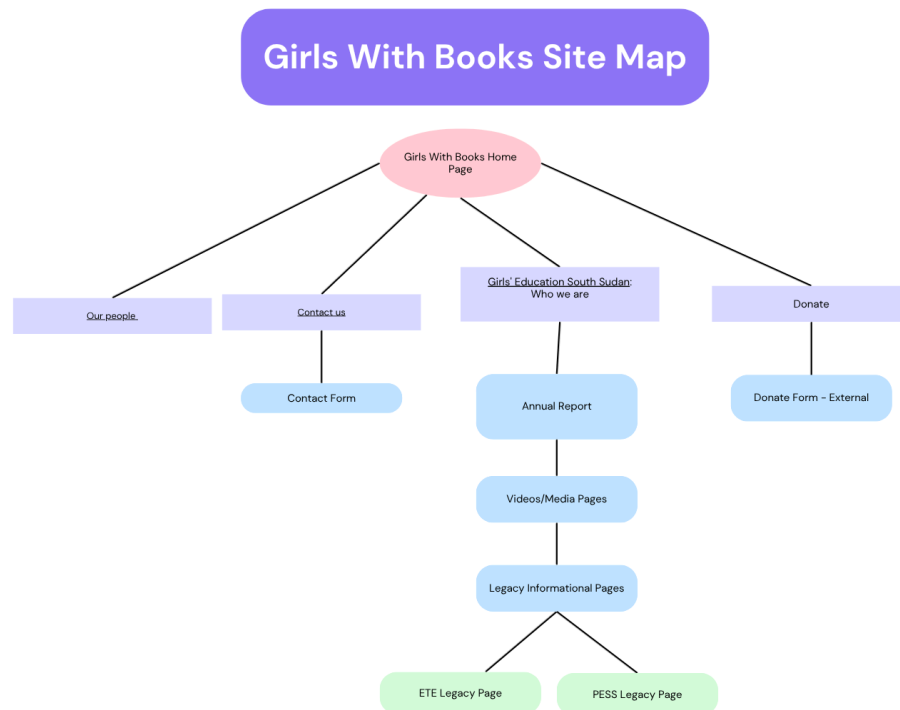


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P.O. Box 27
Boulder, CO 80501

Educating South Sudanese girls and young women to become powerful and positive change makers in South Sudan.

Site Map of Entire Site (up to 5 pages)

We originally made the first site map with what we understood it to be intuitively, so we ended up redesigning it in the Draft 1 by creating the Education page as its own navigation item separate from the home page. So it is not changed in this draft, but rather the first one should have had the home page and the who we are page as one entity.



Site Recode

Home Page: <https://trle5720.github.io/atls-4519/final-project/pages/home.html>

- Consistent Navigation (WCAG 2.1 Guideline 2.4.5 Multiple Ways, 2.4.4 Link Purpose)
 - Previously: The initial page upon entering the website is the true home page, but the “Home” and logo links in the Navigation bar direct the user to a different page that is about Girl’s Education. So, once you leave the home page initially, there is no way to get back to that page without manually changing the URL to go to the “/home” route.
 - Fixed: Made the home links route to the correct home page, and created a new navigation item for the “Girl’s Education” page
- Page Titles (WCAG 2.1 Guideline 2.4.2 Page Titled)

- Previously: The page titles did not match the correct page, the home page and the Girl's education page sharing the same page title and being structured inconsistently as the other pages. For example, those two pages were titled "Girls With Books | Girls' Education South Sudan", while all the other pages were reversed to be in the order of "Our People | Girls With Books".
- Fixed: On all the pages, we fixed the page titles to accurately reflect the content of each page in consistent manners
- Color Contrast (WCAG 2.1 Guideline 1.4.3 Contrast (Minimum))
 - Previously: The original site had contrast errors between white text and the clashing background that was an image of students. The page was pretty overwhelming with the color clashing.
 - Fixed: Our redesign takes away the clashing background and ensures there is no low contrast errors by making the background white and the text dark colors that meet the ratio.
- Image Alternative Texts (Guideline 1,1 Text Alternatives: 1.1.1 Non-text Content A)
 - Previously: The original site had unhelpful alternative text for images, such as "IMG_9842.jpg" that doesn't provide any helpful information about the images.
 - Fixed: The recoded pages have helpful alternative texts for all the images that describe what is in it.
- Heading Structure (WCAG 2.1 Guideline 2.4.6 Headings and Labels AA)
 - Previously: There was no heading hierarchy, the whole site just utilized h3 and normal text
 - Fixed: Follow the correct heading structure by starting with h1 and continuing to follow the hierarchy by moving down to h2, rather than just jump straight to h3.

Donate Page: <https://trle5720.github.io/atls-4519/final-project/pages/donate.html>

- Form (WCAG Guideline: 2.1.1 Keyboard A)
 - Previously: The form on the Donate page does not allow for radio selection for the payment form section using only a keyboard. It just skips over it when trying to Tab into it
 - Fixed: Made the whole form accessible by ensuring all fields can be selected using keyboard as well as adding in fieldset and legend attributes to make the sections read out loud and clearer for screen readers

Remote User Testing Report

Goals

- Can the user distinguish page content and read text clearly without confusion caused by low color contrast or color reliance?
- Can the user navigate between pages without error, including finding their way back to the home page?

- Does the participant experience confusion due to mismatched page titles when using multiple browser tabs?
- Does the heading structure support the participant's reading and orientation strategies?

Tasks

1. Read homepage content: You've discovered Girls With Books! and want to know more about the organization. Please explore the homepage and tell me what you find out.
2. Navigate to home: You've been browsing the Donate page. Please go back to the home page.
3. Scan the home page for key topics: As quickly as you can, please scan the page and tell me what major topics or sections you can identify.
4. Make a donation: You'd like to support the organization. Please navigate to the Donate page and try to complete the donation form using your keyboard.

User 1: Emma

User profile

- Age range: 18-25
- Disabilities: red-green colorblindness and dyslexia
- Assistive technology: browser zoom and occasional screen reader use
- Device: Google Chrome on Macbook
- Experience: Moderate web user who is somewhat familiar with nonprofit websites. This was her first visit to girlswithbooks.org.

Task Performance

Task	Before Redesign	After Redesign
Read homepage content	Partial – Emma was able to read the homepage content, though she found it difficult with the lack of contrast between the text and background image. She said she found the background image distracting.	Success – Emma was able to fully read and understand the homepage. She did not have difficulty due to lack of contrast.
Navigate to home	Fail – After leaving the home page, she was unable to find her way back. She was only able to get to the Girls' Education page.	Success – Emma found it much easier to navigate back home when the donate page remained in the same tab and she could see in the navigation bar how to get back to the home page.

Scan page for key topics	Success – She was able to visually scan the home page visually and find the key topics.	Success – She was still able to visually scan the home page visually and find the key topics.
Make a donation	Fail – Emma was unable to make a donation when only using her keyboard. She was unable to select a donation amount.	Success – She was able to fill out the form only using her keyboard.

Project Draft 3:

- We were unable to get cooperation from the owner or employee from the site, so we conducted a secondary user remote test

User 2: Andrew

User profile

- Age range: 35-44
- Disabilities: motor disability affecting upper limbs
- Assistive technology: standard keyboard with sticky keys, screen reader (on throughout testing)
- Device: FireFox on Windows PC
- Experience: Experienced web user. This was his first visit to girlswithbooks.org.

Task Performance

Task	Before Redesign	After Redesign
Read homepage content	Partial – Andrew’s screen reader struggled to convey the page structure meaningfully without a correct heading hierarchy. The poor contrast made it hard for him to place where he was on the page.	Success – Andrew was able to read the homepage content with his screen reader correctly identifying new sections.
Navigate to home	Fail – He was unable to navigate back to the home page. Upon clicking the home button, he was routed to the Girls’ Education page which he found very frustrating.	Success – Andrew was able to quickly navigate back to home and knew immediately that he was on the right page with his screen reader correctly identifying the home page.

Scan page for key topics	Fail – His screen reader was unable to identify the headings and he was unable to use his keyboard to quickly navigate to new sections due to a lack of h1/h2 anchors.	Success – He was able to quickly scan the home page for section titles with the new hierarchy.
Make a donation	Fail – Andrew was unable to make a donation while using his keyboard. He was unable to select a donation amount.	Success – Andrew was able to complete all required steps to make a donation.

Summary

Task	Emma - Before/After Redesign	Andrew Before/After Redesign	WCAG Guideline
Read homepage content	Partial → Success	Partial → Success	1.4.3 (AA), 2.4.6 (AA)
Navigate to home	Fail → Success	Fail → Success	2.4.2 (A)
Scan page for key topics	Success → Success	Fail → Success	2.1.1 (A), 2.4.6 (AA)
Make a donation	Fail → Success	Fail → Success	2.1.1 (A)

The redesign successfully resolved all critical accessibility barriers identified during initial testing. Both participants were able to complete every task after the changes were applied.

Before the redesign, the most prominent accessibility barriers were the broken keyboard accessibility on the donation form (both users failed this task entirely), the absent heading hierarchy (which prevented Andrew's screen reader and keyboard from scanning the page), low color contrast on the homepage, and broken navigation routing. The original site produced at least one failure or partial result for every task across both users.

After the redesign, all eight task attempts resulted in a full success. The fixes that had the broadest impact were the heading structure correction, the form keyboard accessibility improvement, the corrected navigation links and page titles which eliminated disorientation across tabs and after page loads, and the removal of the low-contrast background image which directly improved readability for Emma. Taken together, the changes brought the site into meaningful compliance with the targeted WCAG guidelines and produced a measurably better experience for users with a varying range of disabilities.

